Anticoagulation in Patients Requiring Antiplatelet Therapy

Objective:

To provide guidance to clinicians on the management of patients with indications for both antiplatelet and anticoagulant drugs.

Background:

Patients who have an indication for oral anticoagulation (OAC) can develop separate conditions that would normally be treated with antiplatelet agents. Atrial fibrillation (AF) and acute coronary syndrome (ACS), with or without requirement for percutaneous coronary intervention (PCI), overlap in approximately 20% of patients. Other indications for anticoagulation, such as venous thromboembolism (VTE), mechanical heart valves, and left ventricular (LV) thrombus may also co-exist with an indication for single or dual antiplatelet therapy (SAPT, DAPT). Low-dose rivaroxaban therapy in combination with acetylsalicylic acid (ASA) has demonstrated benefit in patients with coronary artery disease with or without peripheral artery disease.

Antithrombotic management of patients requiring combined anticoagulant and antiplatelet therapy necessitates an **individualized consideration of thrombotic risk, including stroke and systemic embolism (SSE), stent thrombosis, and ACS, as well as bleeding risk** (see **Tables 1 and 2**). The rationale for combining DAPT (P2Y12 inhibitor + ASA) with OAC is to reduce thromboembolic events, cardiovascular morbidity, and mortality. However, it comes with an increased risk of major bleeding.

Randomized trials have demonstrated reduced bleeding risk with the use of a dual pathway inhibition strategy (P2Y12 inhibitor + DOAC or warfarin) over triple therapy (DAPT + OAC). Although no increased thrombotic risk has been noted with this strategy, individual studies have been underpowered for thrombotic outcomes. Atrial fibrillation was the primary indication for anticoagulation for most included patients, and data specific to other indications for anticoagulation is limited.

Management strategy for patients with AF and ACS/PCI requiring antiplatelet therapy

A treatment algorithm recommended by the Canadian Cardiovascular Society and Canadian Association of Interventional Cardiology is shown in **Figure 1**. They recommend that patients with AF + PCI or ACS should first be stratified according to their risk of stroke or systemic embolism based on CHADS2 plus age. Those at low risk (CHADS2 =0 age <65) have no indication for anticoagulation so may be managed in the same way as those without AF. For those with CHADS2 > 0 or age >= 65, triple therapy with OAC, ASA, and clopidogrel should be employed, then ASA should be discontinued between 1 and 30 days after initiation, taking into consideration the risk of stent thrombosis and bleeding. Those at high risk of stent thrombosis (see **Table 1**) and at low bleeding risk may experience more benefit from longer durations of triple therapy, while those at low risk of stent thrombosis and high risk of bleeding (see **Table 2)** may benefit more from a dual pathway strategy beginning 1 day after PCI.

Ticagrelor and prasugrel are not recommended in combination with OAC as few patients in the relevant trials were treated with these two P2Y12 inhibitors. Of note, the studied and recommended OACs are rivaroxaban 15 mg daily (10 mg daily if CrCl 30-50 mL/min), apixaban 5 mg BID (2.5 mg BID if age $\geq$ 80, Cr > 133umol/L, or weight < 60 kg are met), dabigatran 150 mg or 110 mg BID, edoxaban 60 mg daily (30 mg daily if CrCl 15-50 mL/min, weight < 60 kg or concurrent use of potent P-glycoprotein inhibitors) or warfarin. If warfarin is used, the recommended INR target is 2.0-2.5.

Figure 1: Antiplatelet strategies in patients with concomitant atrial fibril
percutaneous coronary intervention (PCI) and/or acute coronary syndr

∗ Patients who do not require OAC for their AF as per the Canadian guidelines for the management of AF would receive dual antiplatelet therapy.

∗∗ ASA duration before switching to dual pathway was an average of 1.6-6.6 days after PCI in the large-scale trials.

Redrawn from Bainey KR, et al. Can J Cardiol 2024;40:160-181.

Table 1: Factors Associated with Increased Thrombotic Risk with PCI

Table 2: Factors Associated with Increased Bleeding Risk

Management strategy for patients with VTE and ACS/PCI requiring antiplatelet therapy

For patients with VTE, oral or parenteral anticoagulation should be prescribed as recommended for those who do not require antiplatelet therapy. For patients in whom extended or indefinite oral anticoagulation is required, use of an agent with the potential for dose reduction can be considered (e.g., rivaroxaban to 10 mg daily or apixaban to 2.5 mg bid when used beyond 6 months).

When patients anticoagulated for VTE undergo PCI with or without ACS, we recommend following the same strategy as for patients whose indication for anticoagulation is AF (see **Table 1 and Table 2**), i.e., a short duration of triple-therapy followed by dual pathway inhibition for up to one year. Of note, patients with a VTE within the last 3 months should receive therapeutic doses of OAC even with concomitant use of antiplatelet agents. Patients requiring elective PCI should have the procedure delayed, if possible, until anticoagulation is no longer required for VTE management or the risk of recurrence with anticoagulant interruption is minimized (see **Clinical Guide: Venous Thromboembolism Duration of Therapy**). If OAC is discontinued, it is recommended to resume the antiplatelet strategy appropriate to the outcome of ACS/PCI (e.g., re-starting ASA to complete one year of DAPT).

Management strategy for patients with LV thrombus and ACS/PCI requiring antiplatelet therapy

Patients with LV thrombus have a high risk of thromboembolic events, and anticoagulation is recommended for them. A meta-analysis of data from small RCTs and larger observational studies suggests similar efficacy and safety of DOACs compared to warfarin for treatment of LV thrombus, albeit with a low degree

scientific statement from the American Heart Association found DOACs to be a reasonable alt
for this indication, especially for patients in whom time in therapeutic range for warfarin is pred

When patients with LV thrombus undergo PCI with or without ACS, we recommend using a similar approach as for patients whose indication for anticoagulation is AF (see **Tables 1 and 2**), i.e., a short duration of triple-therapy followed by dual pathway inhibition for up to one year if anticoagulation is continued. Of note, in most studies for this indication, DOACs were used at full therapeutic dose without dose reduction even with concomitant use of antiplatelet therapy. Optimal duration of OAC in these patients is unclear, and decisions regarding continuation of OAC should be made on a case-by-case basis; in general, OAC may be discontinued after 3 months if there has been resolution of the LV thrombus. Once OAC is discontinued, it is recommended to resume the antiplatelet strategy appropriate to the outcome of ACS/PCI (e.g., re-starting ASA to complete one year of DAPT). Based on current evidence, anticoagulation is not recommended for patients at risk for but without confirmed LV thrombus, such as those with anterior STEMI.

Management strategy for patients with mechanical heart valves requiring PCI and antiplatelet therapy

In patients with a mechanical heart valve, warfarin is specifically recommended over DOACs due to the higher risk of valve thrombosis and SSE with DOACs. Therapeutic warfarin should be resumed as soon as possible after PCI to mitigate these risks with an INR target dictated by the type of valve. When patients with mechanical heart valves undergo PCI with or without ACS, we recommend using the same approach as for patients whose indication for anticoagulation is AF (see **Tables 1 and 2**), i.e., a short duration of triple-therapy followed by dual pathway inhibition for up to one year.

Low dose rivaroxaban and ASA in patients with stable cardiovascular disease or lower extremity revascularization

The role of low dose rivaroxaban 2.5 mg PO BID for patients with stable CAD (COMPASS trial) and peripheral arterial disease (VOYAGER PAD trial) has been explored. These trials found a benefit for rivaroxaban in reducing thromboembolic events (including acute limb ischemia and need for major amputation for vascular causes) with an increased risk of major bleeding. Of note, COMPASS excluded patients on DAPT, and VOYAGER-PAD excluded patients with an expected duration of DAPT > 6 months. Given these results, the addition of rivaroxaban 2.5 mg PO BID to low-dose ASA can be considered in patients with stable atherosclerotic vascular disease with a high thrombotic risk that is thought to outweigh the bleeding risks associated with this combination (see **Clinical Guide: Peripheral Arterial Disease**).

Long-term use of an antiplatelet agent with concomitant therapeutic dose anticoagulation

Limited data is available to guide clinicians on which patients could benefit from receiving indefinite use of a combination of low dose ASA with a full dose anticoagulant. Guidelines suggest anticoagulation alone in most patients who have an indication for an anticoagulant (most commonly atrial fibrillation) and are at least 1 year from an ACS or PCI. A 2025 meta-analysis of randomized controlled trials found no difference in the primary efficacy outcomes (MI, stroke and systemic embolism, death), and a higher risk of major bleeding in patients on OAC plus SAPT with an absolute risk difference of 2.4%. This included four trials that enrolled patients with AF and stable CAD, with a mix of patients on DOACs, warfarin, ASA, and P2Y12 inhibitors. Potential scenarios of when to consider long-term use of ASA with a full dose anticoagulant include antiphospholipid antibody
and mechanical heart valves.

APS: This syndrome is associated with a significantly elevated risk of both arterial and venous thrombosis. Patients with this syndrome who develop arterial thrombosis have a higher risk of episodes. A 2022 meta-analysis of randomized controlled trials comparing warfarin to DOACs risk of arterial thromboembolism in patients assigned to DOAC. Guidelines suggest that the addition of ASA to warfarin may be considered in patients with APS who experience a recurrent arterial or venous thrombosis despite already being on warfarin and in the target INR range (INR 2-3). It may also be considered in patients with an index arterial thromboembolic event. It is recommended that all such patients be referred to a thrombosis specialist or hematologist. See **Clinical Guide: Thrombophilia: Antiphospholipid Antibody Syndrome**).

Mechanical Heart Valves: Older randomized studies reported lower rates of systemic embolism and potentially lower rates of death when ASA was added to warfarin in patients with mechanical heart valves. Experts suggest using this combination in this scenario, especially if the patient is at low risk of bleeding or at a high thrombotic risk. For some valves (e.g., On-X valve in the aortic position), a lower INR target is recommended specifically when low dose ASA is used in combination with warfarin. See **Clinical Guide: Bioprosthetic and Mechanical Heart Valves: Anticoagulant Therapy.**

Other relevant

- Acetylsalicylic Acid (ASA)
- Apixaban (Eliquis®)
- Bioprosthetic and Mechanical Heart Valves: Anticoagulant Therapy
- Clopidogrel (Plavix®)
- Dabigatran (Pradaxa®)
- Deep Vein Thrombosis (DVT): Treatment
- Mechanical and Bioprosthetic Heart Valves: Anticoagulant Therapy
- Peripheral Arterial Disease
- Prasugrel (Effient®)
- Pulmonary Embolism (PE): Treatment
- Rivaroxaban (Xarelto®)
- Stroke Prevention in Atrial Fibrillation
- Thrombophilia: Antiphospholipid Antibody Syndrome
- Ticagrelor (Brilinta®)
- Venous Thromboembolism: Duration of Therapy
- Warfarin

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